

Projection Phrase 2 Documentation

Feature: Recipe Comments System

1. Feature Overview:

For Phrase 2 of the Dishbook project, I implemented a comment system for the recipe pages. This feature allows logged-in users to post comments directly on individual recipes and see their comments appear on the recipe page without reloading the page.

The feature has the following functionality:

- Users can write a comment on any recipe page posted on web server
- Comments are stored permanently in the database
- Comments are displayed below the recipe content
- New comments do not require reloading pages
- Default comments is “no comments yet” in the comment section before any comments are added, but once there is at least one comment the default message disappears without refreshing page.

User interaction flow: User first navigates to a recipe page, writes a comment in the text box, clicks “Post Comment” button to publish it. JavaScript then sends the comment to the server using a POST request. Django also stores the comment in the database. The server returns the comment as JSON, and then JavaScript dynamically updates the page to display the comment immediately.

This feature improves user engagement and encourage user to leave feedback after using/following the recipe to make a dish.

2. Data Model:

I added a new Django model called Comment for this feature in model.py. I have user (foreign key to User model), recipe (foreign key to Recipe model), content (a text field storing the comment text), and created_at (date time field when comment is created) in the Comment model. The model stores each comment and links it to the user who wrote it and the recipe it belongs to. The way I design the model allows multiple comments per recipe and tracking which user wrote each comment and the posting time of each comment.

3. Files Changed:

Models.py: I added Comment model and defined its relationships to User and Recipe.

Views.py: I added logic for handling POST requests in recipe_detail method and returned JSON response for JavaScript.

Urls.py: I update recipe route to use recipe_detail view

Recipe.html: I added comment display section, and the placeholder for the “No comments yet”.

Main.js: I implemented JavaScript fetch request for posting comments, updates DOM dynamically after successful submission, removes placeholder text when first comment is added.

4. End to End feature walk through:

1. Model/Data layer: when a user submits a comment, a new comment object is created and saved to the database.
2. View/server logic: recipe_detail view will receive POST request, extracts comment content, create a Comment object, returns JSON data containing comment info.
3. Template/UI layer: recipe.html will display existing comments from the database and include a comment form.
4. JavaScript behavior: JavaScript (main.js that I just added) will intercepts form submission, sends POST request using fetch, receive JSON response, dynamically updates the page without reloading.

5. JavaScript interaction:

the JavaScript behavior I added acts:

- Prevents page reload on form submission
- Sends POST request using fetch
- Receives JSON response from Django
- Insert new comments and remove comment placeholder

It will benefit from JavaScript rather than server side because without JavaScript, users would need to reload the page after every comment and interaction would feel slow and outdated.

6. AI Usage:

I used ChatGPT to help me generate some ideas of what the new feature could be and I also used it for debugging Django view errors. I ran into an issue where my Django fetch request returning HTML instead of JSON and my comments won't show up on the UI. Some prompts I used to let ChatGPT help me debug are “why is my Django fetch request returning HTML instead of JSON”, “This is the CSRF token error in console when I submit a comment, can you tell me what the error mean?”, etc.

7. Human Judgment

Several places I changed after interacting with AI are 1) routing confusion and 2) debugging JSON vs HTML response. I corrected the URL routing to ensure POST requests reached the correct handler and the JavaScript parsing errors after checking them with AI. It redirected me to the right place for debugging and explain what the error means in the console when I try posting comments but they are not showing on the screen

8. Testing and edge cases

First, I tested comments normally by posting some shorter comments and longer comments, and multiple comments on the same receipt and see if they will show up without reload. Some edge cases I tested are submitting comments that's only made of symbols (&*^\$%...), submitting empty comment, and verifying the correct comment order after submitting multiple comments (the new comments appear at the top). I learned that it's important to identify edge cases and we should try our best to find a case to make our program "fail" and debug step by step afterwards.

9. Screenshots:

This is what the recipe page looks like before I post any comments. Notice the "No Comments yet" placeholder exists at this point.

Ada Lovelace | [Sign out](#)

Tomato Soup

By [Alan Turing](#)

vegetarian | comfort

A warm and comforting tomato soup for chilly days.

Prep time: 10 minutes

Cook time: 20 minutes

Total time: 30 minutes

Serves: 2

Ingredients

- No ingredients listed

Steps

- Heat the tomatoes and broth in a pot.
- Blend until smooth and simmer before serving.

Comments

No comments yet

This is what the page looks like after I submit one comment. Placeholder disappeared, and the new comment is time-stamped and shows the name of user who posted it:

Tomato Soup

By [Alan Turing](#)

vegetarian | comfort

A warm and comforting tomato soup for chilly days.

Prep time: 10 minutes

Cook time: 20 minutes

Total time: 30 minutes

Serves: 2

Ingredients

- No ingredients listed

Steps

- Heat the tomatoes and broth in a pot.
- Blend until smooth and simmer before serving.

Comments

Write a comment...

Post Comment

alovelace

I love this receipe! It's delicious and easy to follow.

2026-04-15 15:01

And if I post another comment, the new comment appears above the old comment:

Cook time: 20 minutes

Total time: 30 minutes

Serves: 2

Ingredients

- No ingredients listed

Steps

- Heat the tomatoes and broth in a pot.
- Blend until smooth and simmer before serving.

Comments

Write a comment...

Post Comment

alovelace
highly recommend to try it at home
2026-04-15 15:02

alovelace
I love this receipe! It's delicious and easy to follow.
2026-04-15 15:01

The comments also can be hold in the comment box before you publish it:

Ingredients

- No ingredients listed

Steps

- Heat the tomatoes and broth in a pot.
- Blend until smooth and simmer before serving.

Comments

comments will show up in the comment box
when you type it

Post Comment

alovelace
highly recommend to try it at home
2026-04-15 15:02

alovelace
I love this receipe! It's delicious and easy to follow.
2026-04-15 15:01